

modul lab

- ① IoT based Smart home application
- ② IoT based Smart gardening
- ③ IoT devices in networking.
- ④ Simulate a multimedia h/w in ept.

Off
Verdict
Q

①

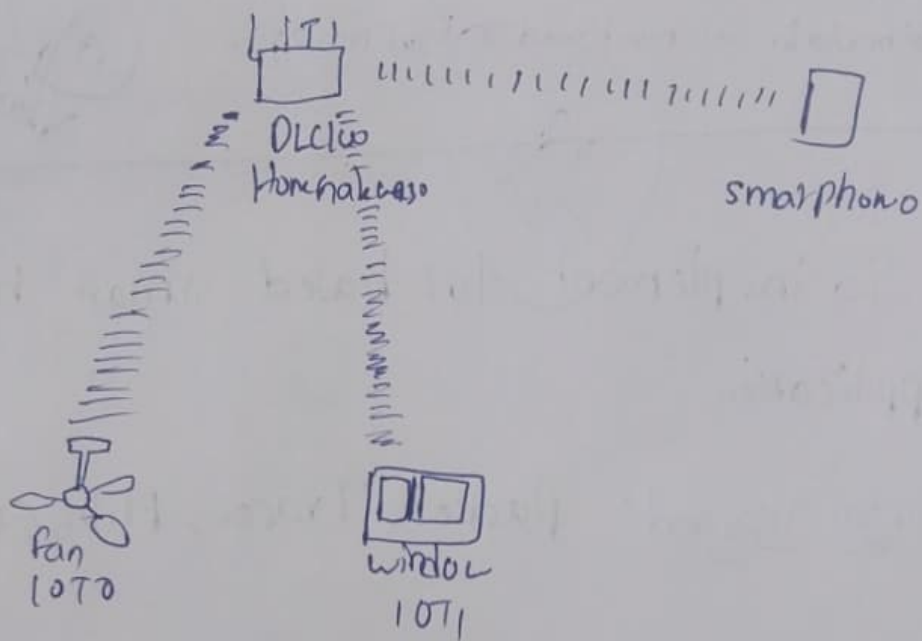
Aim: To implement IoT based smart home application

Apparatus required: Packet Tracer, Hubs, Connector.

Procedure:

- open Cisco Packet Tracer, select Home Gateway from the IoT device
- Add smart home device, such as smart light, fan, door and window
- Connect the IoT device wirelessly to the Home Gateway
- Add a smart phone, now connect the smartphone to the same network
- Configure the IoT device, with the Home Gateway
- Open the IoT Control application on the smart phone
- Turn on/off the smart light and fan

Test the smart door and window.



Result

The IoT based smart Home was successfully designed and simulated using Cisco Packet Tracer.

②

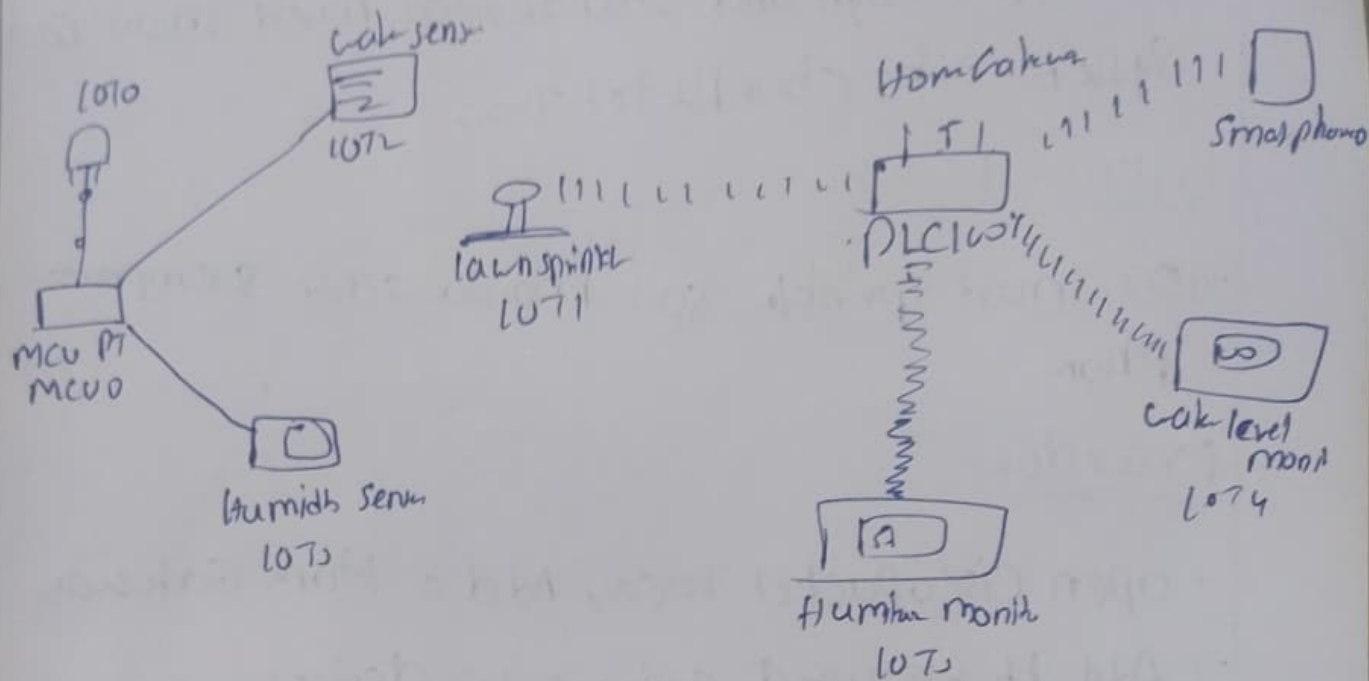
Aim: To design and simulate an IoT based smart Gardening System using Cisco Packet Tracer.

Apparatus Required

- IoT smart sprinkler, soil moisture sensor, ~~Temperature~~ smart Phone

Procedure

- open Cisco Packet Tracer, Add a Home Gateway.
- Add the required gardening IoT devices
- Place a soil moisture sensor in the garden
- Add a smart sprinkler / Water Pump. Connect the devices to the Home Gateway.
- Connect a smart phone to the network, configure the sensor and sprinkler
- Set the sprinkler to operate when the soil moisture becomes low.
- Run the simulation



Result

The IoT-based smart gardening system was successfully simulated. and the sprinkler was controlled according to the soil condition.

③ Aim:- To study and demonstrate the connection and communication of IoT device in a computer network using Cisco Packet Tracer.

Apparatus Required:-

Router/Wireless Router, Switch, PC, smart phone, IoT device, Ethernet cables

Procedure

- open Cisco Packet Tracer, place a router or wireless router.
- Add a switch and connect it to the router.
- Add a PC and connect it to the switch. Add IoT device such as smart lights and fans.
- now connect the IoT devices to the wireless connection.
- Configure IP addresses for the network device, configure the IoT devices to connect to the network.
- use the Command Prompt on the PC. use the ping command to test connectivity.

④ Aim:

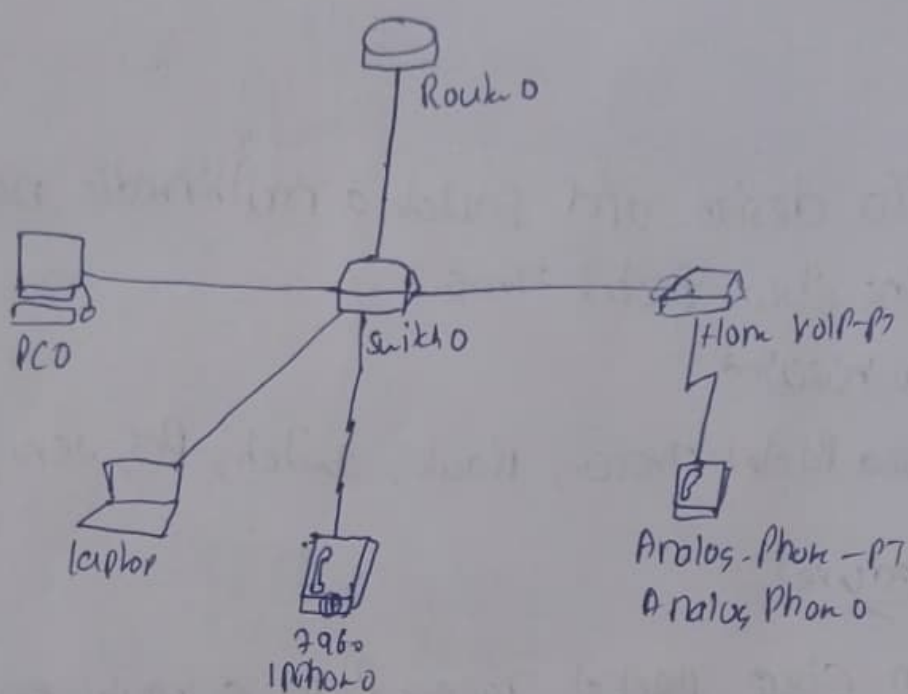
To design and simulate a multimedia network using Cisco Packet Tracer.

Apparatus required:

Cisco Packet Tracer, Router, Switch, PC, Server, IP Phone

Procedure:

- 1 open Cisco Packet Tracer, Place a router and switch.
- 2 Connect the switch to the router. Add PC and multimedia device.
- 3 Connect all devices using Ethernet cables.
- 4 Assign IP addresses to the devices and configure the router and network.
- 5 Test connectivity using the ping command.
- 6 Use simulation mode to observe packet capture the transmission of multimedia related data through the network.



Result:-

The multimedia network was successfully designed and simulated using Cisco Packet Tracer.